

TOYOOKA, Japan

WMO: 477470

Lat: 35.5358N

Lon: 134.8222E

Elev: 19

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	29.7	21.4	16.0	38.6	23.8	18.0	39.1	13.9	40.0	12.1	39.8	3.0	130	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.1	94.7	78.0	92.4	77.5	90.0	76.8	79.7	90.0	78.8	88.6	77.9	86.9	7.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	141.4	83.5	76.2	137.0	83.0	75.3	132.9	82.5	43.2	90.3	42.3	88.7	41.4	87.4	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.4	11.6	9.9	DB	24.4	99.1	2.2	1.6	22.8	100.2	21.5	101.1	20.3	102.0	18.7	103.1
			WB	23.7	81.5	2.4	1.2	21.9	82.4	20.5	83.1	19.2	83.8	17.4	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	38.5	39.4	45.5	55.4	65.0	72.2	80.1	82.1	74.0	63.4	53.0	43.5	(6)
	DBStd	15.86	4.05	4.97	5.95	6.42	5.05	4.63	4.55	4.03	5.42	5.21	5.39	5.26	(7)
	HDD50	1083	356	300	167	21	0	0	0	0	0	0	28	211	(8)
	HDD65	3623	821	717	603	293	63	3	0	0	2	93	361	667	(9)
	CDD50	4538	0	4	28	183	466	666	932	997	719	416	118	9	(10)
	CDD65	1602	0	0	0	5	64	219	467	532	271	43	1	0	(11)
	CDH74	13731	0	0	1	84	549	1352	4306	5466	1787	183	3	0	(12)
	CDH80	5337	0	0	0	13	111	371	1797	2447	571	27	0	0	(13)
(14) Wind	WSAvg	3.7	3.2	3.5	4.1	4.4	4.5	3.9	3.9	4.0	3.6	3.2	3.1	3.3	(14)
(15) Precipitation	PrecAvg	79.5	9.4	7.0	5.6	3.9	5.2	5.9	6.8	5.6	9.3	6.4	6.3	8.2	(15)
	PrecMax	98.9	15.1	11.1	8.4	6.5	17.2	11.2	16.9	13.2	20.4	19.3	13.8	17.0	(16)
	PrecMin	58.4	4.3	3.5	2.2	1.7	1.5	3.0	1.9	0.5	2.6	2.0	2.7	3.9	(17)
	PrecStd	8.3	2.6	2.0	1.8	1.5	3.0	2.1	3.5	3.1	4.5	3.5	2.6	3.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.4	64.2	71.6	81.5	86.7	91.0	96.8	97.8	93.6	83.5	73.0	63.2	(19)
		MCWB	48.1	54.9	55.7	61.6	66.4	72.9	78.2	78.2	76.7	71.1	61.2	54.8	(20)
	2%	DB	51.7	56.8	65.9	75.9	83.1	87.1	93.7	95.5	89.8	78.5	68.5	58.4	(21)
		MCWB	44.9	49.0	53.8	59.0	64.3	72.2	78.0	78.3	75.7	67.8	59.8	51.3	(22)
	5%	DB	48.5	51.9	61.6	72.2	80.1	84.2	91.6	93.5	86.2	75.4	65.2	55.0	(23)
		MCWB	43.4	45.8	51.9	58.0	63.4	70.8	77.6	77.9	74.7	66.1	58.2	49.2	(24)
	10%	DB	45.5	48.1	57.5	68.3	76.7	81.4	89.2	91.2	82.9	72.5	62.3	52.1	(25)
		MCWB	41.3	43.0	49.8	56.5	62.7	70.1	76.9	77.6	73.1	64.3	56.2	47.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	57.3	59.2	66.5	71.4	77.1	80.7	81.4	78.7	73.7	64.7	57.1	(27)
		MCDB	54.7	63.9	65.1	72.8	78.7	86.0	91.4	92.6	88.0	80.7	68.5	58.8	(28)
	2%	WB	46.3	50.2	56.1	62.7	68.6	75.3	79.4	80.0	77.2	70.5	61.9	53.0	(29)
		MCDB	50.2	55.0	62.4	71.4	75.9	82.5	89.7	90.6	85.9	75.9	66.2	56.5	(30)
	5%	WB	43.7	46.5	53.1	60.2	66.3	73.8	78.4	79.2	76.0	68.0	59.6	50.1	(31)
		MCDB	47.6	51.1	60.1	68.5	74.8	80.5	88.3	89.2	83.6	72.4	63.8	54.2	(32)
	10%	WB	41.8	43.3	50.2	58.2	64.6	72.1	77.6	78.3	74.7	65.9	57.3	47.7	(33)
		MCDB	45.2	47.7	56.8	65.8	73.5	78.7	86.7	87.7	81.1	70.5	61.5	51.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	13.4	17.7	20.9	20.0	15.9	14.8	16.1	14.7	16.1	15.3	13.1	(35)
		MCDBR	17.7	22.0	27.0	28.4	27.2	21.2	19.2	19.8	18.8	19.4	20.2	18.8	(36)
	5% WB	MCWBR	11.3	14.1	15.6	13.6	11.1	8.2	6.4	6.1	7.0	9.2	11.3	11.6	(37)
		MCWBR	16.4	20.2	22.1	21.2	19.7	15.6	17.3	17.3	15.8	15.1	16.7	17.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.432	0.466	0.530	0.552	0.553	0.575	0.575	0.544	0.494	0.455	0.434	0.426	(40)
		taud	2.025	1.961	1.832	1.838	1.895	1.915	1.973	2.054	2.149	2.157	2.117	2.061	(41)
	Ebn at Noon		230	239	234	238	240	234	233	238	243	240	229	223	(42)
		Edh at Noon	42	50	62	65	62	61	57	52	45	41	38	39	(43)
(44) All-Sky Solar Radiation	RadAvg		542	762	1112	1465	1628	1509	1509	1566	1213	975	674	500	(44)
	RadStd		43	85	101	130	163	120	229	178	111	90	56	45	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.94	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page